



Mahindra University Hyderabad
École Centrale School of Engineering
Minor I

Program: B. Tech. Branch: CM Year: II Semester: Fall
Subject: Real Analysis (MA2104)

Date: 11/09/2025
Time Duration: 1.5 Hours

Start Time: 10.00 AM
Max. Marks: 20

Instructions:

1. All questions are compulsory.
2. Please start each answer on a separate page.

Q 1:

4 marks

Classify the sets below as open, closed or neither open nor closed. Justify.

- (i) $(0, 1) \cup (5, 8]$ (ii) $\bigcup_{n=1}^{\infty} (a_n, b_n)$ (iii) $\{1, \frac{1}{2}, \frac{2}{3}, \dots, \frac{n}{n+1}, \dots\}$ (iv) $\{\pm 1, \pm 2, \pm 3, \dots\}$

Q 2:

4 marks

Show that the set of all integers and the set of all rationals are countable sets.

Q 3:

5 marks

Explain the construction of a Cantor set and show that the Cantor set is uncountable.

P.T.O.

Q 4:

3 marks

Let (x_n) and (y_n) be two sequences converges to x and y respectively. Then show that the sequence $(x_n y_n)$ converges to xy .

Q 5:

4 marks

Justify your answers in the following:

- (i) Is there exists a set $E \subset \mathbb{R}^1$ such that E and its complement E^c are dense in $\mathbb{R}^1$.
 - (ii) Is the countable intersection of open sets open?
 - (iii) Is $A = \{0, 1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, \dots\}$ a perfect set?
 - (iv) Let $f : X \longrightarrow Y$ be a function and $A \subset X$ and $B \subset X$. Is $f(A \cap B) = f(A) \cap f(B)$?
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